

User Guide

RM126x AT Interface Application

Version 1.9

Revision History

Version	Date	Notes	Contributor(s)	Approver
1.0	09 August 2023	Initial Release	Greg Leach	Senthooran Ragavan
1.1	07 September 2023	Updated for finalisation of P2P feature. Added further P2P parameterisation to ATS section. Added read-only P2P parameterisation to ATI section. Restricted available data rates for AS923 and US915 in P2P mode in ATS section. Added link to firmware update guide to Appendix A – References. Removed DR6 and DR7 from supported P2P data rates in ATS section.	Greg Leach	Senthooran Ragavan
1.2	24 October 2023	Updated following current optimization. Added details of the necessity for a wakeup message to allow the host interface UART to recover from sleep mode in section 5.2.11. Removed AT+READ and AT+RESEND commands for long term sleep current optimisation approach. Renamed P2P Listen Frequency parameter to P2P Listen Interval.	Greg Leach	Senthooran Ragavan
1.3	6 December 2023	Moved Public Mode parameter from location 610 to new location 709. This is to ensure the parameter is set to the intended default value of Public. Added separate asynchronous P2P receive message to section 5.2. Updated asynchronous LoRaWAN and P2P receive messages to include RSSI and SNR values. Added details of minimum value for UART Idle Time parameter.	Greg Leach	Senthooran Ragavan
1.4	11 April 2024	Ezurio rebranding. Added description for new Slots per Window parameter (ATS 710). Updated description of P2P functionality to account for the optimisation made in the GA1.4 release. Added description of Max Slots per Window parameter (ATI 4006).	Sue White Leo Genuardi	Senthooran Ragavan
1.5	20 May 2024	Updated P2P address range and broadcast address details following P2P optimisation (GA1.4). Corrected width of AT%S parameters 635, 636 and 637.	Greg Leach	Senthooran Ragavan
1.6	6 August 2024	Added quotes around message parameter for AT+SEND and AT+P2PT commands.	Greg Leach	Senthooran Ragavan

1.7	30 April 2025	Updated for GA1.5 release. Added details of ADRI, ADRX and ADRJ asynchronous events. These are issued in event of ADR related parameters changing during (J)oining, by e(X)ternal MAC commands from the LNS, or (I)nternally as the device works through its internal ADR mechanism. Removed references to NZ915 region. Added details of Preamble Count ATS parameter. This is used to synchronise with host MCU devices waking from sleep mode where an initial number of module characters may be lost. Added details of Gen App Key AT%<i>s</i> parameter. Added details of write-only access rights to AT%<i>S</i> parameters App Key, ABP & P2P Network Session Key and ABP & P2P Application Session Key. Added details of AT+BAT command. This is used to set device battery level details in DevStatusReq MAC command responses. Added details of Static ADR Datarate (713) and Static ADR TX Power (714) ATS parameters. Added details of AT+DUTY command. This is used to indicate whether a packet length is permissible with the remaining duty cycle time. Reduced number of possible values in AT<i>I</i> parameters 42 (Application State) and 2016 (LoRa Radio activity) to avoid duplication of values. Added details of AT+SUPC, AT+SUPI and AT+SUPS commands and associated asynchronous responses for module supply voltage measurement. Added details of AT+SLEEP command for finer grain control of module sleep mode. Added details of AT+CRX and AT+CTX command and associated AT%<i>S</i> parameters for manufacturing test continuous transmit and receive modes. Removed ATS parameters 612 (Request Count) & 613 (RX1 Delay) to allow control by expected conformance test behaviour or MAC commands. Added details of TX Power Limit (715) ATS parameter to allow a maximum transmit power level to be defined. P2P Broadcast address reverted to 64.	Rikki Horrigan Seokwoo Yoon Greg Leach	Jonathan Kaye
1.8	31 May 2025	Updated for GA1.6 release. Added note to ATZ command to ensure AT&W is invoked prior to resetting the device when Activation Mode is set to ABP. This ensures the uplink frame count is stored and continuity maintained with the Network Server. Added details of asynchronous message issued when Activation Mode is set to ABP following reset of the module.	Greg Leach	Jonathan Kaye
1.9	15 October 2025	Corrected available values for AT<i>I</i> 2016 parameter LoRa Radio Activity.	Greg Leach	Jonathan Kaye

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1 Overview

This document is a user guide for the AT Interface application written for the RM126x [A] LoRa modules. It exposes an industry standard AT command/response protocol. This protocol instructs the module to transmit and receive in LoRaWAN Classes A, B or C, and provides access to a proprietary Peer to Peer protocol using the LoRa physical layer. It also provides commands to read and write to GPIO pins.

The latest version of the AT Interface application is always available from the RM126x firmware Github repository [B].

Note: New feature requests are encouraged via Github pull requests or Sales channels.

2 Application Loading Instructions

Refer to [C] for details of programming the AT application firmware.

3 Operation

3.1 Configuration

The RM126x AT Interface application is configured via a plurality of non-volatile parameters, these are divided into numeric and string-based datatypes. Writable numeric parameters are configured using the **ATS** AT command, and writable string parameters are configured using the **AT%S** AT command.

Note: Non-volatile parameters must be saved following alteration using the **AT&W** AT command. The new values are not applied until the next power-cycle. This can be performed using the **ATZ** AT command which triggers a warm reset of the module.

Read-only parameter data providing runtime information, and fixed module parameters, can be interrogated via the **ATI** command. Factory default non-volatile data can be restored at any time using the **AT&F** AT command.

3.2 Modes

The RM126x can be configured to operate in either LoRaWAN or Peer to Peer modes.

LoRaWAN based modes are dependent upon LoRaWAN infrastructure and can be described as follows.

3.2.1 Class A

In this mode, the RM126x initiates communication with the network. Only upon reception of an uplink message can the network send downlink data. Class A operation is foundational, with Class B and C also implementing its features.

3.2.2 Class B

Class B operation offers the network more opportunities to send downlink data to the remote device. In this mode, the device regularly listens for messages from the network. The frequency at which this occurs is defined by the Ping Slot Periodicity parameter, with lower values resulting in more frequent listen operations and higher power consumption.

3.2.3 Class C

Devices in Class C mode can be reached at any time by the network. When not sending uplink data, the device listens continuously for downlink data from the network. Due to the radio being continuously active, Class C operation tends to be suitable only for line powered devices.

3.2.4 Peer to Peer (P2P)

Peer to Peer operation obviates the need for LoRaWAN infrastructure. In this mode, devices communicate directly with each other using LoRa modulation. An addressing system is used to filter incoming messages to determine when a response needs to be sent to a request.

Refer to [E] for further details.

3.3 Data Flow Control

The host that is driving the UART interface must strictly adhere to RTS/CTS handshaking to ensure that data buffering and management are not compromised. If the module de-asserts its RTS line, the host stops sending data as soon as possible and conversely, if the host de-asserts its RTS line, the module stops sending data to it.

Note: The UART host should expect asynchronous (unsolicited and arriving anytime) responses which contain downlink, link check reports and other data. These could be interleaved with normal responses. For this reason, every response type in section 5.2 has a unique identifier which allows the host to demultiplex them appropriately.

4 AT Commands

These are text commands starting with the character sequence AT and terminated by a `\r` character (ASCII code 0x0D). Commands are not case sensitive and have zero or more parameters. Multiple parameters are separated by the comma (,) character. Some commands tolerate empty fields (two consecutive commas) and provide a default value. If more parameters are supplied than those specified, then the extra parameters are either silently ignored or result in a syntax error response.

The AT commands are described in this section in alphabetical order and are grouped by functional area. Also listed are responses to these commands which are described in the next chapter.

Many commands take parameters which are either integer values, strings or hex strings:

- Integer values – Can be entered as binary or in hexadecimal using the syntax 0xhh..hh where hh is a hexadecimal number.
- Strings – Textual data that must be enclosed within inverted commas
- Hex strings – Only contain the letters 0-9, A-F and a-f and shall be exactly an even number long. Otherwise they are treated as syntax errors

Note: Spaces are shown between the AT commands and the parameters accepted. The spaces are not mandatory and included for visual clarity.

4.1 [Empty Line]

Command	Empty line with any amount of whitespace characters
Possible Responses	OK

4.2 AT

Command	AT
Description	No action performed other than to send the OK response
Possible Responses	OK

4.3 Parameter Access Commands

4.3.1 AT%S

Command	AT%S n="string" AT%S n? AT%S n=?
Description	These commands are respectively used to set, get, and retrieve the range of valid lengths of any writable string parameter. The parameter ID is identified by the integer value n. n="string" - When setting values, "string" is the new value; the double quotes are mandatory. Non-printable characters can be embedded in the new value. These should be escaped using the three-character \hh sequence where hh is the ASCII value of the character in hexadecimal. For example, to null terminate the string "Hello", it is entered as "Hello\00"

n? and **n=?** – For these variants, the returned value is enclosed in \n and \r and are sent before the OK. The two integer values returned by **n=?** are separated by a comma.

Note: Modified values are not retained over a power-cycle or a warm reset triggered using the **ATZ** command. Refer to the AT&W command to make modified values permanent.

The following writable string parameters are defined.

ID	Description
0	<i>Device Name</i> The length is between 1 and 20 alpha-numeric characters.
500	<i>App Key</i> 128-bit LoRaWAN App Key. The length is 32 hexadecimal characters. This parameter is write only.
501	<i>Dev EUI</i> 64-bit LoRaWAN Dev EUI. The length is 16 hexadecimal characters.
502	<i>Join EUI</i> 64-bit LoRaWAN Dev EUI. The length is 16 hexadecimal characters.
635	<i>ABP & P2P Device Address</i> 32-bit ABP & P2P Device Address The length is 8 hexadecimal characters. For P2P devices, this value must be the same for all devices in the network. This parameter is write only.
636	<i>ABP & P2P Network Session Key</i> 128-bit ABP & P2P Network Session Key. The length is 32 hexadecimal characters. For P2P devices, this value must be the same for all devices in the network. This parameter is write only.
637	<i>ABP & P2P Application Session Key</i> 128-bit ABP & P2P Application Session Key The length is 32 hexadecimal characters. For P2P devices, this value must be the same for all devices in the network. This parameter is write only.
712	<i>Gen App Key</i> 128-bit Remote Multicast App Key The length is 32 hexadecimal characters. This parameter is write only.

Possible Responses
OK
ERROR

4.3.2 AT&F

Command	AT&F
Description	Sets all writable parameters to default values <i>and</i> clears the trusted device bond database then performs a warm reset.
Possible Responses	OK (after warm reset) ERROR

4.3.3 AT&W

Command	AT&W
Description	Saves all writable parameters to non-volatile memory.
Possible Responses	OK ERROR

4.3.4 AT+I

Command	AT+I n														
Description	Queries read-only parameter data identified by the integer argument n. Values returned are either integer or string data. Returned values are preceded by a \n character and terminated with a \r character. The following information is returned with identifier 'n' as stated. <table> <tr> <th>ID</th><th>Description</th></tr> <tr> <td>0</td><td> <i>Device Type</i> The unique device type. Returned as "RM1261" for RM1261 variants and "RM1262" for RM1262 variants. </td></tr> <tr> <td>3</td><td> <i>Application Firmware Version</i> The firmware version of the module. This is returned in the format WWW.XXX.YYY.ZZZ, with the fields being defined as follows. - WWW is the unique product number, with 127 indicating RM1261 and 128 indicating RM1262 - XXX is the top-level revision of the Silabs SDK in use - YYY is the major release number - ZZZ is the minor release number </td></tr> <tr> <td>42</td><td> <i>Application state</i> This provides a high-level indication of the current application state. Possible values are as follows. 0: Idle 1: Joining 2: Joined 3: Disconnected </td></tr> <tr> <td>1002</td><td> <i>Minimum baud rate</i> This is the minimum configurable baud rate of the AT Interface UART. </td></tr> <tr> <td>1003</td><td> <i>Maximum baud rate</i> This is the maximum configurable baud rate of the AT Interface UART. </td></tr> <tr> <td>2000</td><td> <i>Reset Reason</i> This provides more details on the reason for the reset. Possible values are as follows: 0: None 1: AT&F issued 2: ATZ issued 17: Hard fault 18: Bus fault 19: Usage fault 33: Brown out 34: Brown out while in low energy mode </td></tr> </table>	ID	Description	0	<i>Device Type</i> The unique device type. Returned as "RM1261" for RM1261 variants and "RM1262" for RM1262 variants.	3	<i>Application Firmware Version</i> The firmware version of the module. This is returned in the format WWW.XXX.YYY.ZZZ , with the fields being defined as follows. - WWW is the unique product number, with 127 indicating RM1261 and 128 indicating RM1262 - XXX is the top-level revision of the Silabs SDK in use - YYY is the major release number - ZZZ is the minor release number	42	<i>Application state</i> This provides a high-level indication of the current application state. Possible values are as follows. 0: Idle 1: Joining 2: Joined 3: Disconnected	1002	<i>Minimum baud rate</i> This is the minimum configurable baud rate of the AT Interface UART.	1003	<i>Maximum baud rate</i> This is the maximum configurable baud rate of the AT Interface UART.	2000	<i>Reset Reason</i> This provides more details on the reason for the reset. Possible values are as follows: 0: None 1: AT&F issued 2: ATZ issued 17: Hard fault 18: Bus fault 19: Usage fault 33: Brown out 34: Brown out while in low energy mode
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2001	<i>Cause of last reset</i>
	This indicates the root cause of the reset.
	Possible values are as follows:
	0: Unknown reset cause 1: Power on reset 2: Software reset 3: Brown out 4: Reset pin 5: Lockup 6: Watchdog
2008	<i>Current TX Power</i>
	This is the TX Power in dBm currently in use.
2016	<i>LoRa Radio activity</i>
	This enumeration is used to indicate the current activity being performed by the LoRa radio.
	Possible values are as follows.
	0 : Idle 1 : Transmitting 2 : Receiving
3000	<i>Bandwidth</i>
	This indicates the LoRa Bandwidth currently in use.
	Possible values are as follows:
	0: LoRa_BW_125 1: LoRa_BW_250 2: LoRa_BW_500 3: LoRa_BW_FSK 4: Unknown
3001	<i>Connection Status</i>
	This indicates the connection status to an LNS.
	Possible values are as follows:
	0: Not Connected 1: Connected
3002	<i>Device Address</i>
	Network assigned device address.
3003	<i>Bootloader Firmware Version Number</i>
	This is the version number of the device Bootloader firmware.
3004	<i>Modulation Type</i>
	This is the modulation type currently in use.
	Possible values are as follows:
	0: LoRa 1: GFSK
3005	<i>Module Type</i>
	This is a single digit code representing the module type.
	Possible values are as follows:
	0: RM1261 1: RM1262
3006	<i>Packets RXed</i>

		Represents the total number of packets received since the last power cycle.
3007	<i>Packets TXed</i>	Represents the total number of packets transmitted since the last power cycle.
3008	<i>RSSI</i>	Represents the last read back RSSI value in dBm. For LoRaWAN, this is updated only after downlink messages are received following joining.
3009	<i>Signal To Noise Ratio</i>	Represents the last read back SNR value in dB. For LoRaWAN, this is updated only after downlink messages are received following joining.
3010	<i>Spreading Factor</i>	This is the LoRa Spreading Factor currently in use. Possible values are as follows: • 5: SF5 • 6: SF6 • 7: SF7 • 8: SF8 • 9: SF9 • 10: SF10 • 11: SF11 • 12: SF12
3011	<i>Data Rate</i>	This is the LoRaWAN Data Rate currently in use for uplink messages. Possible values are as follows: • 0: DR0 • 1: DR1 • 2: DR2 • 3: DR3 • 4: DR4 • 5: DR5 • 6: DR6 • 7: DR7
4000	<i>P2P Calculated Beacon TOA</i>	This is the time in milliseconds the synchronising beacon occupies within a P2P window.
4001	<i>P2P Beacon Settle Time</i>	This is the time in milliseconds following completion of a beacon send before the initial receive or send slot is permitted.
4002	<i>P2P Slot TOA</i>	This is the time in milliseconds the maximum allowable packet size for the configured P2P data rate occupies within a P2P window.
4003	<i>P2P Slot Packet Size</i>	This is the maximum number of bytes that can be sent during a send slot for the configured P2P data rate.
4004	<i>P2P Slot Delay</i>	This is the time in milliseconds between successive send and/or receive slots.
4005	<i>P2P Minimum Window Length</i>	This is the time in seconds for the minimum allowable window length.

4006	<i>P2P Maximum Slots per Window</i>
This is the maximum number of slots available for P2P messages for the selected region.	
4007	<i>CRX DevEUI</i>
This is the currently configured DevEUI for Continuous Receive tests.	

OK
ERROR

4.3.5 ATS

Command	ATS n=m ATS n? ATS n=?												
Description	These commands are respectively used to set, get and retrieve the range of valid values, for any writable numeric parameter. The parameter is identified by the integer value n. When setting, m is the new value. For the 'n?' and 'n=?' variants the returned value will be enclosed in \n and \r and will be sent before the OK. The max and min integer values returned by 'n=?' are separated by a comma. Note when setting the value that it will not be retained over a power-cycle or a warm reset triggered using the ATZ command. Refer to the AT&W command to make all changed values permanent. The following writable numeric parameters are defined. <table> <tr> <th>Parameter</th><th>Description</th></tr> <tr> <td>100</td><td> <i>Start-up Flags</i> Bit 0: Set to Trigger joining upon start-up </td></tr> <tr> <td>117</td><td> <i>SIO used to indicate network connection</i> If set to -1, this SIO is disconnected. When set to a valid SIO number, this SIO is set high when joined to a network and low otherwise. </td></tr> <tr> <td>213</td><td> <i>UART Idle Time in milliseconds for low power UART operation</i> If no UART activity is detected for this length of time the UART will be automatically closed. If there is incoming data over the air that needs to be conveyed to the host, the UART is automatically opened. A minimum time of 500ms is allowed for this parameter, values less than this result in the UART low power operation being disabled. </td></tr> <tr> <td>217</td><td> <i>Core Voltage in Millivolts</i> This is the operating voltage the module is intended to operate at. This is used to define the reference used by AD routines to properly scale input voltages against the reference value defined here. This defaults to 3300mV. </td></tr> <tr> <td>302</td><td> <i>UART Baud rate</i> This specifies the baud rate to use for host UART commands and data transfer. Defaults to 115200bps. </td></tr> </table>	Parameter	Description	100	<i>Start-up Flags</i> Bit 0: Set to Trigger joining upon start-up	117	<i>SIO used to indicate network connection</i> If set to -1, this SIO is disconnected. When set to a valid SIO number, this SIO is set high when joined to a network and low otherwise.	213	<i>UART Idle Time in milliseconds for low power UART operation</i> If no UART activity is detected for this length of time the UART will be automatically closed. If there is incoming data over the air that needs to be conveyed to the host, the UART is automatically opened. A minimum time of 500ms is allowed for this parameter, values less than this result in the UART low power operation being disabled.	217	<i>Core Voltage in Millivolts</i> This is the operating voltage the module is intended to operate at. This is used to define the reference used by AD routines to properly scale input voltages against the reference value defined here. This defaults to 3300mV.	302	<i>UART Baud rate</i> This specifies the baud rate to use for host UART commands and data transfer. Defaults to 115200bps.
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600	<i>ADR Enabled</i> Enables and disables ADR. Used for mobile devices where ADR cannot reliably be used. LinkADR requests received from the LNS will still be processed but changes to the datarate and TX Power inhibited. Possible values are as follows: 0: ADR disabled 1: ADR enabled When disabled, the data rate is set using ATS parameter 713 and the TX Power using ATS parameter 714. Defaults to ADR enabled.
601	<i>Beacon Frequency</i> Sets the beaconing frequency for Class B devices.
602	<i>Activation Mode</i> Selects the activation mechanism used. Possible values are as follows: 0: ABP 1: OTAA Defaults to OTAA.
603	<i>Device Class</i> Selects the method of LoRa operation. Possible values are as follows: 0: Class A 1: Class B 2: Class C 3: P2P Defaults to Class A.
604	<i>Confirmed Packets</i> <i>Used to enable and disable packet confirmation from the LNS.</i> Possible values are as follows: 0: Unconfirmed Packets 1: Confirmed Packets Defaults to Confirmed Packets.
605	<i>Confirmed Retry Count</i> Sets the number of retries attempted when a confirmed packet is not acknowledged by the LNS. Defaults to 3, allowable values between 0 and 10.
606	<i>Join Delay</i> Sets the number of milliseconds waited following request for a join attempt before the join procedure is started.
607	<i>Join Jitter</i> Sets the maximum number of milliseconds to be either added or subtracted from the Join Delay before the join procedure is started. A random number is generated using this value as a seed to ensure a group of devices do not all attempt a join procedure at once.
609	<i>Ping Slot Periodicity</i> For Class B devices, sets the number of ping slots following an uplink. 0: Ping slot every 1s 1: Ping slot every 2s

2: Ping slot every 4s

3: Ping slot every 8s

4: Ping slot every 16s

5: Ping slot every 32s

6: Ping slot every 60s

7: Ping slot every 128s

Decreasing values of Ping Slot Periodicity increase the chances of reception of incoming messages but also decrease battery life.

611

Region

Sets the region of operation.

Possible values are as follows:

- 0: Unconfigured

RM1261 only:

- 1: EU_EU868
- 2: UK_EU863_870
- 8: TAIWAN_AS923_1
- 9: JAPAN_AS923_1
- 11: INDIA_IN865_867

RM1262 only:

- 3: USA_CANADA_US902_928
- 4: AUSTRALIA_AU915_928
- 5: AUSTRALIA_AS923_1
- 6: NEW_ZEALAND_AS923_1

617

Sub-band

For 915 region devices, this sets the 8 channel sub-band in use by the device.

Possible values are as follows:

- 0: Channels 0 - 63
- 1: Channels 0 - 7
- 2: Channels 8 - 15
- 3: Channels 16 - 23
- 4: Channels 24 - 31
- 5: Channels 32 - 39
- 6: Channels 40 - 47
- 7: Channels 48 - 55
- 8: Channels 56 - 63

629

Application Port

Sets the port used for uplink data.

Allowable values are between 1 and 198.

633

Class C Scan Time

For Class C mode, determines how often checks are made for incoming data. As this decreases responsiveness improves but with increasing power consumption.

700

P2P Device Address

This is the unique address of the P2P device in a network. Devices in the network should be addressed in ascending order, starting from 0. The device at address 0 acts as the synchronisation point for all other devices in the network.

701

P2P Network Size

This is the number of P2P devices in the network, with allowable values between 2 and 63.

702	<i>P2P Window Length</i>	This is the duration in seconds between synchronisation events being issued by the P2P device residing at Device Address 0. This needs to be long enough to cater for each device in the network to occupy one transmit slot such that each peer has an opportunity to send a message within each window. If the P2P Window Length is set to too short a value, an appropriate error code is returned.
703	<i>P2P Data Rate</i>	This is the data rate used by devices to send messages within P2P windows. Increasing values allow for increased message size at the expense of range. Allowable values as follows. 0: DR0 (Not available for AS923, Australia AU915) 1: DR1 (Not available for AS923, Australia AU915) 2: DR2 3: DR3 4: DR4 (Not available for US/Canada US915) 5: DR5 (Not available for US/Canada US915)
704	<i>P2P Listen Duration</i>	Before joining a P2P network, devices with Device Addresses above 0 must first detect a synchronisation event from Device 0. This parameter defines the time spent in seconds listening for events. Increasing times improve likelihood of detecting an event at the expense of battery life. Setting this parameter to 0 results in the peer adopting Class C listening behaviour, at the expense of battery life.
705	<i>P2P Listen Interval</i>	This parameter determines time in seconds between Listen Durations being triggered. Decreasing times improve the likelihood of detecting a synchronisation event at the expense of battery life. Note this value must be longer than the P2P Listen Duration. Setting this parameter to 0 results in the peer adopting Class C listening behaviour, at the expense of battery life.
706	<i>P2P Beacon Data rate</i>	This is the data rate used for the P2P synchronising beacon. As the data rate decreases, range increases, but the minimum P2P Window Length also increases. This results in less throughput and longer response times. Allowable values as follows: 0: DR0 (Not available for AS923, Australia AU915) 1: DR1 (Not available for AS923, Australia AU915) 2: DR2 3: DR3 4: DR4 (Not available for US/Canada US915) 5: DR5 (Not available for US/Canada US915)
707	<i>P2P Packet Size</i>	This is the longest packet size intended for sending across the P2P network. This represents the longest packet size, with shorter packet lengths also being permitted for sending. This parameter should be used to optimise the network Minimum Window Length.

708	<i>P2P TX Power</i> This is the transmit power in dBm the peer will operate at. It is recommended to limit this value for peers operating in close proximity to each other. Maximum allowable values as follows. AU915: 22 AU923: 14 EU868: 13 IN865: 13 JP923: 12 NZ923: 14 TW923: 14 US915: 22
709	<i>Public Mode</i> Enables and disables public network operation. Possible values are as follows: 0: Private 1: Public Defaults to Public.
710	<i>Slots Per Window</i> This is the number of send & receive slots assigned when in P2P mode. The maximum value applied depends on the number of channels available, which is region specific, and the number of devices in the network. The minimum value applied should equal the number of devices in the network. A value of 0 will automatically apply the maximum number of slots for the currently configured region and network size. Any other value between the minimum and maximum must be divisible by the number of devices in the network. Below are the number of channels available in the different regions while in P2P mode. AU915: 63 AU923: 32 EU868: 19 IN865: 16 JP923: 16 NZ923: 32 TW923: 9 US915: 63 Defaults to 0.
711	<i>Preamble Count</i> For devices where Low Power UART operation is enabled, this parameter allows a number of preamble characters to be sent following asynchronous wake-up. These characters can be safely discarded by the host and allow wake-up behaviour on its side to be safely performed without data loss. Each preamble character is set to a hexadecimal value of 0xFF, with an STX character (0x02) being used to indicate the start of asynchronous message data. A maximum of 10 preamble characters can be sent. Defaults to 0.

713 *Static ADR Datarate*

Applicable only when parameter 600 (ADR Enabled) is set to 0, sets the fixed Datarate to be used by the device.

Allowable values as follows.

AU915: 0 .. 6

AU923: 0 .. 6

EU868: 0 .. 7

IN865: 0 .. 5

JP923: 0 .. 5

NZ923: 0 .. 7

TW923: 2 .. 5

US915: 0 .. 4

Defaults to 2.

714 *Static ADR TX Power*

Applicable only when parameter 600 (ADR Enabled) is set to 0, sets the fixed TX Power to be used by the device.

AU915: 0 .. 22

AU923: 0 .. 14

EU868: 0 .. 13

IN865: 0 .. 13

JP923: 0 .. 12

NZ923: 0 .. 14

TW923: 0 .. 14

US915: 0 .. 22

Note: The TX Power Limit parameter will take precedence over the Static ADR TX Power parameter. It's recommended to disable the TX Power Limit by setting to -1 if Static ADR is being used.

Defaults to 0.

715 *TX Power Limit*

Sets the maximum TX Power usable by the module. This will be used during join requests and can only be reduced by the LNS following successful joining. Maximum allowable values are shown below.

AU915: 0 .. 22

AU923: 0 .. 14

EU868: 0 .. 13

IN865: 0 .. 13

JP923: 0 .. 12

NZ923: 0 .. 14

TW923: 0 .. 14

US915: 0 .. 22

The minimum value permitted in all cases is 0. Setting this value to -1 disables the TX Power Limit.

Defaults to -1.

OK

ERROR

4.4 I2C Commands

I2C is a two-wire serial protocol. It facilitates communication between host microcontrollers and external peripheral devices. The I2C connections are defined as follows.

- SDA: This is a bidirectional line used by the host device to send data to external peripheral devices, and for the peripheral device to send data to the host.
- SCL: This is permanently configured as an output by the host microcontroller. It is used to clock individual data bits into the controller and peripheral devices.

Note: Prior to usage of the I2C commands, the appropriate SIO configuration is required using the **AT+SIOC** commands.

Note: Refer to [D] for examples of I2C commands being used.

4.4.1 AT+I2R

Command	AT+I2R hexDataString, outLength	
Description	This command is used to perform an I2C device read. Data is returned in hexadecimal format.	
	hexDataString	This is the hexadecimal data string sent to the device. This should include the device address byte and associated read command data. Data should be sent in hexadecimal format. For example, an address value of 1 and a data byte of 254 would be sent as 01FE.
	outLength	This is the number of bytes to read back from the device
Possible Responses	OK ERROR	

4.4.2 AT+I2W

Command	AT+I2W hexDataString	
Description	This command is used to perform an I2C device write.	
	hexDataString	This is the hexadecimal data string sent to the device. This follows the same format as read command data.
Possible Responses	OK ERROR	

4.5 LoRaWAN Class A Commands

This section describes available LoRaWAN Class A associated commands.

4.5.1 AT+DROP

Command	AT+DROP	
Description	This command is used to drop a connection to a network, or to cancel any pending join requests.	
Possible Responses	OK ERROR	

4.5.2 AT+JOIN

Command	AT+JOIN
Description	This command is used to begin joining to a network.
Possible Responses	OK ERROR

4.6 LoRaWAN Class B and C Commands

The following describe commands applicable only to Class B and C modes of operation.

Multicast groups are sub-networks residing within a LoRaWAN application. The devices in the multicast group can be accessed simultaneously by the network with downlink messages.

4.6.1 AT+MCGA

Command	AT+MCGA groupId, groupAddress, nsKey						
Description	This command is used to create a multicast group. <table> <tr> <td>groupId</td><td>This is the 8-bit id used to identify the group, allowable values between 0 and 3.</td></tr> <tr> <td>groupAddress</td><td>This is the 32-bit address assigned to the group.</td></tr> <tr> <td>nsKey</td><td>This is the 128-bit Network Session Key, sent in hexadecimal format.</td></tr> </table>	groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.	groupAddress	This is the 32-bit address assigned to the group.	nsKey	This is the 128-bit Network Session Key, sent in hexadecimal format.
groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.						
groupAddress	This is the 32-bit address assigned to the group.						
nsKey	This is the 128-bit Network Session Key, sent in hexadecimal format.						
Possible Responses	OK ERROR						

4.6.2 AT+MCGE

Command	AT+MCGE groupId		
Description	This command is used to end a multicast group session. <table> <tr> <td>groupId</td><td>This is the 8-bit id used to identify the group, allowable values between 0 and 3.</td></tr> </table>	groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.
groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.		
Possible Responses	OK ERROR		

4.6.3 AT+MCGL

Command	AT+MCGL groupId		
Description	This command is used to list the details of a multicast group. <table> <tr> <td>groupId</td><td>This is the 8-bit id used to identify the group, allowable values between 0 and 3.</td></tr> </table>	groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.
groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.		
Possible Responses	OK ERROR		

4.6.4 AT+MCGR

Command	AT+MCGL groupId		
Description	This command is used to remove a multicast group. <table> <tr> <td>groupId</td><td>This is the 8-bit id used to identify the group, allowable values between 0 and 3.</td></tr> </table>	groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.
groupId	This is the 8-bit id used to identify the group, allowable values between 0 and 3.		
Possible Responses	OK ERROR		

4.6.5 AT+MCGS

Command	AT+MCGL groupId
Description	This command is used to start a multicast session. Note a group must first be configured using AT+MCGA for the session to be started. groupId This is the 8-bit id used to identify the group, allowable values between 0 and 3.
Possible Responses	OK ERROR

4.7 LoRaWAN Common Commands

The following commands apply to all LoRaWAN classes.

4.7.1 AT+DUTY

Command	AT+DUTY
Description	The DUTY command accepts a single byte parameter representing the length of next payload to be sent. This length is checked against the maximum allowable length according to remaining duty cycle time, and returns an appropriate error code if the packet would violate duty cycle restrictions. payloadLength The payload length to check against remaining duty cycle time.
Possible Responses	OK ERROR

4.7.2 AT+LINKC

Command	AT+LINKC
Description	The LINKC command performs a link check. This is used to determine how many gateways are in the vicinity of the module and the signal quality received by the gateway with the best margin. Refer to JOIN [OK] When Activation Mode is configured as ABP, the JOIN asynchronous message is issued following module reset. Note join parameters are omitted due to no join sequence being performed. LC for asynchronous responses.
Possible Responses	OK ERROR

4.7.3 AT+SEND

Command	AT+SEND "hexDataString"
Description	This command is used to uplink a data packet. hexDataString This is the hexadecimal data string sent to the network within quote characters. Refer to TX for asynchronous responses. For Class A devices where downlink messages can only be sent following an uplink, refer to RX (LoRaWAN) for asynchronous messages associated with downlink events.

Possible Responses	OK
	ERROR

4.7.4 AT+TIME

Command	AT+TIME
Description	This command is used to request the time from the network server. Refer to TIME for asynchronous responses.
Possible Responses	OK
	ERROR

4.8 P2P Commands

This section describes the commands available when the module is configured for P2P operation. Refer to [E] for further details.

4.8.1 AT+P2PE

Command	AT+P2PE
Description	This command ends a P2P session.
Possible Responses	OK
	ERROR

4.8.2 AT+P2PS

Command	AT+P2PS
Description	This command is used to start a P2P session. For devices at Device Address 0, this will start the broadcast of synchronization events. For devices at Device Addresses beyond 0, this will commence listening for synchronization events.
Possible Responses	OK
	ERROR

4.8.3 AT+P2PT

Command	AT+P2PT id, "data"
Description	This command is used to send data to a peer. <i>id</i> is the address of the peer device, between 0 and 62, to send the message to. A value of 64 is a broadcast message that will be sent to all devices on the network. <i>data</i> is the data in hexadecimal format to be sent within quote characters.
Possible Responses	OK
	ERROR

4.9 SIO Commands

AT Commands with the SIO extension are used to interact with single device SIOs.

4.9.1 AT+SIOC

Command	AT+SIOC <i>sionumber</i>, <i>function</i>, <i>subfunction</i>
Description	The module Signal Input/Output (SIO) pins may have their functionality configured at run-time using this command. <i>sionumber</i> is a value in the range 1 to N which identifies the signal pin number <i>function</i> is the SIO function required <i>subfunction</i> is the SIO function context specific feature that is required. Note: Refer to [D] for details of the features supported by each RM126x SIO. Refer to [D] for details of the function and sub-function codes.
Possible Responses	OK ERROR

4.9.2 AT+SIOR

Command	AT+SIOR <i>sionumber</i>
Description	SIOs configured for Digital Input or AD operation can have their state queried with this command. <i>sionumber</i> is a value in the range 1 to N which identifies the signal pin number If configured as a Digital Input, 0 or 1 are returned to indicate a low or high logic level. For an SIO configured for AD operation, the returned value represents the number of counts read by the RM126x 12-bit AD converter. This is referenced to the supply voltage, as configured via writable numeric parameter 217. To convert this value to voltage data, the following formula should be used: $V_{out} = (4096 / V_{supply}) \times (returned\ AD\ counts)$ The integer value returned has a starting \n character and ending \r character. Note: Refer to [D] for examples of the single SIO input variants being used.
Possible Responses	OK ERROR

4.9.3 AT+SIOV

Command	AT+SIOV <i>sionumber</i>, <i>val</i>
Description	SIOs configured for Digital Output, Frequency or PWM operation can have their state set using this command. <i>sionumber</i> is a value in the range 1 to N which identifies the signal pin number <i>val</i> is the value to apply to the SIO If configured as a Digital Output, a value of 0 sets a logic low level and a value of 1 a logic high level. If configured as a Frequency pin, values of up to 1000000 are permitted here to set the corresponding frequency in Hz on the SIO. If configured as a PWM pin, values of 0 to 100 are permitted here to set the corresponding duty cycle percentage on the SIO. Note that the PWM frequency is fixed to 10kHz.
	Note: Refer to [D] for examples of the single SIO output variants being used.
Possible Responses	OK ERROR

4.10 SPI Commands

SPI is a minimum three-wire serial protocol. It facilitates communication between host microcontrollers and external peripheral devices. The SPI connections are defined as follows.

- CIP0: This is the Controller In Peripheral Out pin, used to clock data into the Controller
- COPI: This is the Controller Out Peripheral In pin, used to clock data into the Peripheral
- CS: Each peripheral device has a Chip Select pin that is driven by the Controller to select the Peripheral device for communication on the SPI bus
- SCK: This is permanently configured as an output by the host microcontroller. It is used to clock individual data bits into the controller and peripheral devices.

Note: Prior to usage of the SPI commands, the appropriate SIO configuration is required using the **AT+SIOC** commands.

Note: CS SIOs should be configured prior to SPI CIP0, COPI and SCK SIOs.

Note: Refer to [D] for examples of the SPI commands being used.

The following commands are used to perform SPI Controller operations on SPI Peripheral devices.

4.10.1 AT+SPR

Command	AT+SPR <i>id</i>, <i>hexDataString</i>, <i>outLength</i>						
Description	This command is used to perform an SPI read operation.						
	<table> <tr> <td><i>id</i></td><td>The SPI device instance configured via the AT+SIOC command with SPI CS1 or CS2 argument, with a value of 1 indicating the device at CS1 and a value of 2 indicating the device at CS2</td></tr> <tr> <td><i>hexDataString</i></td><td>This is the hexadecimal data sent to the device</td></tr> <tr> <td><i>outLength</i></td><td>This is the number of bytes to read back from the device.</td></tr> </table>	<i>id</i>	The SPI device instance configured via the AT+SIOC command with SPI CS1 or CS2 argument, with a value of 1 indicating the device at CS1 and a value of 2 indicating the device at CS2	<i>hexDataString</i>	This is the hexadecimal data sent to the device	<i>outLength</i>	This is the number of bytes to read back from the device.
<i>id</i>	The SPI device instance configured via the AT+SIOC command with SPI CS1 or CS2 argument, with a value of 1 indicating the device at CS1 and a value of 2 indicating the device at CS2						
<i>hexDataString</i>	This is the hexadecimal data sent to the device						
<i>outLength</i>	This is the number of bytes to read back from the device.						
Possible Responses	OK ERROR						

4.10.2 AT+SPW

Command	AT+SPW id, hexDataString	
Description	This command is used to perform an SPI write operation.	
	Id	The SPI device instance configured via the AT+SIOC command with SPI CS1 or CS2 argument, with a value of 1 indicating the device at CS1 and a value of 2 indicating the device at CS2
	hexDataString	This is the hexadecimal data sent to the device.
Possible Responses	OK ERROR	

4.11 Module Management Commands

4.11.1 ATZ

Command	ATZ	
Description	Restart the module by performing a warm reset.	
	Note when Activation Mode is configured as ABP, parameters should be stored using the AT&W command before resetting the module. This will ensure the uplink frame counter, which is not stored every time an uplink is performed, is saved and continuity ensured with the Network Server.	
Possible Responses	OK	

4.11.2 AT+CRX

Command	AT+CRX startTest, "ctxDevEUI"	
Description	The CRX command sets the device into Continuous Receive mode. This is intended for manufacturing test use only. startTest This is set to 1 to start the CRX test. Setting to 0 stops the test. ctxDevEUI Only required when the CRX test is being started. This is the 8-byte DevEUI of the device in Continuous Transmit mode the device being set to Continuous Receive mode will listen for. The DevEUI is sent in ASCII format within quotes and occupies 16 characters. E.g. a DevEUI of 0x1234567812345678 is sent as "1234567812345678".	
Possible Responses	OK	
	ERROR	

4.11.3 AT+CTX

Command	AT+CTX	
Description	The CRX command sets the device into Continuous Transmit mode. This is intended for manufacturing test use only. When started, the transmitting device sends its DevEUI in a LoRa packet every 5s at 1dBm transmit power. A frequency of 902.5kHz is always used, with Spreading Factor 10 and Coding Rate of 4/5 also being fixed for each message transmitted.	
Possible Responses	OK	
	ERROR	

4.11.4 AT+TEMP

Command	AT+TEMP	
Description	This command returns the current module temperature in degrees C.	
Possible Responses	OK	

4.12 Module Power Management Commands

4.12.1 AT+BAT

Command	AT+BAT batteryStatus	
Description	The BAT command sets the device battery status returned as part of responses to DevStatusReq messages from the LNS. A single byte parameter is sent in the command to set the required battery status. batteryStatus The single byte battery status. Defined values as follows. 0 – The device is connected to an external power source. 1..254 – The battery level, where 1 is the minimum and 254 the maximum. 255 – Default value, the device was not able to measure the battery level.	
Possible Responses	OK	
	ERROR	

4.12.2 AT+SLEEP

Command	AT+SLEEP
Description	The SLEEP command overrides the timeout time associated with the Low Power UART (refer to ATS parameter 213) and puts the module immediately to sleep.
Possible Responses	OK ERROR

4.12.3 AT+SUPC

Command	AT+SUPC
Description	The SUPC command cancels a scheduled module supply voltage measurement request queued using AT+SUPS.
Possible Responses	OK ERROR

4.12.4 AT+SUPI

Command	AT+SUPI
Description	The SUPI command performs an immediate measurement of the module supply voltage.
Possible Responses	OK ERROR

4.12.5 AT+SUPS

Command	AT+SUPS
Description	The SUPS command schedules a module supply voltage measurement. The measurement is performed upon commencement of the next uplink message. Voltage drop is most severe here due to the current consumption of the LoRa radio during transmit operations. Refer to SUP for asynchronous responses.
Possible Responses	OK ERROR

5 Responses

To simplify reception of messages in the receiving device, each message starts with a `\n` character and ends with a `\r` character, and may contain additional embedded `\n` characters, where `\n` is the linefeed character with ASCII code 0x0A and `\r` is the carriage return characters with ASCII code 0x0D.

After stripping the `\n` start character, each response will start with a unique 2-character sequence to help the host decode the response quicker in a stateless manner.

Some responses are synchronous which mean they are used to terminate a command so that the command parser can process more commands.

5.1 Synchronous and Terminating

When a host receives these responses, it can issue new commands and expect them to be processed immediately.

5.1.1 ERROR

ERROR `nn`

A command was not successfully actioned and 'nn' is an error code. Error Codes are as follows.

01	Invalid SReg number
02	Value supplied is out of range
05	Syntax Error
09	Invalid Address has been supplied
14	Command cannot be processed in current state
15	Unknown Command
33	Value supplied is not valid
46	GPIO specified is not available
47	Too few parameters supplied
48	Too many parameters supplied
49	Hex String is not valid
50	Save Fail
51	Restore Fail
68	Newline character was expected but was not found
71	I2C Pin Already Defined
72	GPIO Already Defined
73	PWM Not Found
74	PWM Channel Overflow
75	Wrong GPIO Type
76	Frequency Not Found
77	Frequency Channel Overflow
78	ADC Not Found
79	AD Channel Overflow
80	LoRa Already Joined
81	LoRa Not Joined
82	LoRa Insufficient Duty
83	LoRa No TX Message
84	LoRa No RX Message
85	LoRa Multicast Group Recall Failed
86	LoRa Multicast Group Create Failed
87	LoRa TX Set Fail
88	LoRa Invalid Class
89	Parameter write only
99	Functionality not coded

5.1.2 OK

OK

A command was successfully processed.

5.2 Asynchronous and Terminating

5.2.1 ADRI, ADRJ, ADRX

ADRI / ADRJ / ADRX DR:[state] *data_rate*, PO:[state] *transmit_power*, NB:[state] *number_of_transmissions*, CP:[state] *channel_plan*, PL:[state] *payload_length*

The ADRI asynchronous response is issued upon an internally performed ADR activity. This occurs when confirmed uplinks are not acknowledged by the LNS for a protracted number of uplinks.

The ADRJ asynchronous response is issued upon joining and provides details of the LoRaWAN parameters adopted following the join procedure.

The ADRX asynchronous response is issued upon processing of an external ADR request from the LNS. The LNS optimizes the end-node data rate, transmit power and channel plan using ADR messages.

Parameters within ADRI, ADRJ and ADRX messages are defined as follows.

state	Each parameter value within ADRI, ADRJ and ADRX messages are prefixed with a state value. This indicates the following. ^: Indicates the parameter was increased during the ADR request or join event. v: Indicates the parameter was decreased during the ADR request or join event. !: Applicable to channel plan only, indicates the parameter has changed during the ADR request or join event. x: Indicates the parameter was unchanged during the ADR request or join event.
data_rate	This is the transmit data rate in use.
transmit_power	This is the transmit power in use in dBm.
number_of_transmissions	This is the number of transmissions (NbTrans) in use.
channel_plan	This is the channel plan in use.
payload_length	This is the maximum payload length in bytes supported by the data rate in use.

5.2.2 CLASS

CLASS *result*

The CLASS asynchronous message is issued when the device successfully changes class. For Class B devices, this is issued when the network is successfully joined, and time synchronization with the network is achieved.

A result of OK indicates a successful transition to the requested class. A result of FAIL indicates the transition to the class was not successful or has been reverted. For P2P devices, this indicates the beacon lock has been lost.

5.2.3 CRX

CRX: R:*rsi*, S:*snr*

The CRX asynchronous message is issued by devices upon reception of messages from devices in Continuous Transmit mode. For this asynchronous message to be issued by the receiving device, the DevEUI of the device in Continuous Transmit mode must be set as a parameter to the Continuous Receive mode command **AT+CRX**. The DevEUI of the transmitting device can be queried at any time in the receiving device using **ATI** parameter 4007.

rsi	This is the RSSI of the last received message from the device in Continuous Transmit mode.
snr	This is the SNR of the last received message from the device in Continuous Receive mode.

5.2.4 JOIN

JOIN [*result*], [TX]: C:*uplink_channel*, F:*uplink_frequency*, DR:*uplink_datarate*, [RX]: W:*window*, C: *downlink_channel*, F:*downlink_frequency*, DR:*downlink_datarate*

When Activation Mode is configured as OTAA, the JOIN asynchronous response is issued upon completion of a join attempt, triggered via the **AT+JOIN** command.

Result	the result of the join attempt and is returned as either OK upon success or FAIL when the attempt is not successful.
--------	--

The TX parameters are returned irrespective of the result of the join attempt and are defined as follows.

uplink_channel	The channel used to send the join request message.
uplink_frequency	The frequency used to send the join request message.
uplink_datarate	The datarate used to send the join request message.

The RX parameters are returned only when the join attempt is successful. The parameters are defined as follows.

Window	The receive window within which the join accept downlink was received, either 1 or 2.
downlink channel	The channel where the join accept downlink was received.
downlink_frequency	The frequency where the join accept downlink was received.
downlink_datarate	The datarate at which the join accept downlink was received.

JOIN [*OK*]

When Activation Mode is configured as ABP, the JOIN asynchronous message is issued following module reset. Note join parameters are omitted due to no join sequence being performed.

5.2.5 LC

LC: [S:*status*], M:*margin*, C:*gateway_count*

The LC asynchronous response is issued upon completion of a link check command, triggered by the **AT+LINKC** command.

status	The result of the link check command, OK if successful, FAIL otherwise. If unsuccessful, other values are unavailable.
margin	The demodulation margin in dB. Values at 0 and close to 0 indicate poor connection quality. Optimal values are at and close to the transmit power of the module. The value used relates to the gateway with the best margin.
gateway_count	The number of gateways that detected the link check request.

5.2.6 RX (LoRaWAN)

RX : W: *window*, P: *port*, C: *channel*, F: *frequency*, DR: *datarate*, R: *rss*i, S: *snr*, *data*

The RX asynchronous message is issued upon reception of data from the LNS. Parameters returned are defined as follows.

window	The receive window where the downlink arrived with possible values as follows:	
	1	RX1
	2	RX2
	B	Class B
	B_MC0	Class B Multicast Group 0
	B_MC1	Class B Multicast Group 1
	B_MC2	Class B Multicast Group 2
	B_MC3	Class B Multicast Group 3
	C	Class C
	C_MC0	Class C Multicast Group 0
	C_MC1	Class C Multicast Group 1
	C_MC2	Class C Multicast Group 2
	C_MC3	Class C Multicast Group 3
port	The port at which the downlink arrived.	
channel	The channel where the downlink arrived.	
frequency	The frequency where the downlink arrived.	
datarate	The data rate at which the downlink arrived.	
rss	The RSSI measured for the received message.	
snr	The SNR measured for the received message.	
data	The downlink data in hexadecimal format	

5.2.7 RX (P2P)

RX : [P2P] S: *source*, R: *rss*i, S: *snr*, *data*

The RX asynchronous message is issued upon reception of data from a P2P peer. Parameters returned are defined as follows.

source	The address of the peer that sent the message.
rss	The RSSI measured for the received message.
snr	The SNR measured for the received message.
data	The message data.

5.2.8 SUP

SUP : *supply_voltage*mV

The SUP asynchronous message is issued upon completion of a scheduled supply voltage measurement via **AT+SUPS**. Parameters returned are defined as follows.

supply_voltage	This is the measured supply voltage in millivolts from the last scheduled supply voltage measurement.
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5.2.9 TIME

TIME [*result*], *year:month:day hour:minutes:seconds*

The TIME asynchronous message is issued upon completion of a network time request started using the **AT+TIME** command. Parameters returned are defined as follows.

result	The result of the operation. If successful, this is returned as OK and the time details are included. If unsuccessful, this is returned as FAIL and no time details are included.
year	The UTC year in 4-digit decimal format.
month	The UTC month in 2-digit decimal format.
day	The UTC day in 2-digit decimal format.
hour	The UTC hour in 2-digit decimal format.
minutes	The UTC minutes in 2-digit decimal format.
seconds	UTC seconds in 2-digit decimal format.

5.2.10 TX

TX [*result*], C:*channel*, F:*frequency*, DR:*datarate*

The TX asynchronous response is issued upon completion of an uplink using the **AT+SEND** command. Parameters returned are defined as follows.

result	The result of the operation with possible values as follows	
	ACK	The message was sent and acknowledged by the LNS
	FAIL	Sending of the message was failed
	SENT	The message was sent successfully but not acknowledged by the LNS
channel	The channel used for the uplink.	
frequency	The frequency used for the uplink.	
datarate	The data rate used for the uplink.	

5.2.11 WAKE

WAKE

The WAKE asynchronous message is issued upon successful wake up of the host UART from low power mode. Reception of this message can be interpreted as an indication from the module that messages can now be received from the host.

6 Low Power UART Operation

The RM126x AT Interface application requires a host to control it by sending AT commands over the UART interface.

The UART interface that is embedded inside the microcontroller at the heart of the Ezurio module consumes about 1 milliamp when it is open.

Ezurio has shown that it is possible to operate the module in **doze** mode so that the total current consumption can be as low as sub 2 microamps.

The AT Interface firmware monitors activity on the UART interface, and if none is seen for the duration set by the UART Idle Time parameter, shuts the UART down automatically.

The UART requires some time to stabilize upon being re-enabled. An initial message should be sent to wake the UART up and will always be responded to with 'WAKE' to indicate the UART is ready to receive data again. Following reception of the initial message, data exchange can be performed as usual.

7 Appendix A – References

Ref	Details
[A]	Ezurio RM126x datasheet https://www.ezurio.com/documentation/datasheet-rm126x-lorawan-module
[B]	RM126x Firmware Github repository https://github.com/LairdCP/RM126x_Firmware
[C]	RM126x Firmware Options and Upgrading application note https://www.ezurio.com/documentation/user-guide-firmware-options-and-upgrading-rm126x-series
[D]	Application Note – Peripheral Interface Guide – Lyra, Lyra 24 & RM126x Series https://www.ezurio.com/documentation/application-note-peripheral-interface-guide-lyra-lyra24-and-rm126x
[E]	Application Note – P2P User Guide – RM126x Series https://www.ezurio.com/documentation/application-note-p2p-user-guide-rm126x-series

8 Appendix B – Definitions, Abbreviations and Acronyms

Term	Definition
ABP	Activation by Personalisation
ACK	Acknowledged
ASCII	American Standard Code for Information Interchange Defined list of numeric codes used to represent textual characters.
CTS	Clear To Send UART flow control line used by a device to indicate it is ready to receive data.
FUOTA	Firmware Update Over The Air Mechanism for transferring firmware update files to an end-node.
GPIO	General Purpose Input Output Microcontroller pins available for function assignment.
LNS	LoRaWAN Network Server
LoRa	Long Range
LoRaWAN	Long Range Wide Area Network
OTAA	Over the Air Activation
RSSI	Received Signal Strength Indication
RTS	Request To Send UART flow control line used to indicate when a device is ready to send data.
SIO	Signal Input/Output The RM126x module GPIOs available for usage.
SNR	Signal to Noise Ratio
TOA	Time on Air
UART	Universal Asynchronous Receiver Transmitter Established technology used to send and receive data in a serial manner
Whitespace	Any non-printable character with an ASCII code of 0x20 and lower is treated as a whitespace character.

9 Additional Information

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